

(1) A liquid binary mixture of benzene (C_6H_6) and toluene (C_7H_8) that contains 55 wt% benzene is put into an evaporator to yield a vapor stream that is 85 wt% benzene and a residual liquid stream that is 10.6 wt% benzene.

- (a) Draw a schematic of the process and label all of the streams. The mass flow rate of the feed stream is $\dot{m}_f$ and the output streams are given by $\dot{m}_v$ and $\dot{m}_l$ for the vapor and liquid streams respectively.
- (b) Find the ratio of the mass flow rates of the vapor output to the liquid output ($\dot{m}_v / \dot{m}_l$).
- (c) Is the problem specified, overspecified, or underspecified?

(2) Murphy 2.39

(3) Murphy 2.58

(4) Murphy 2.60

(5) Murphy 2.64